week4

PBL:Hugging Face NLP Project
Team3
track 3 & session 3
Sara Hamada, Yadi Feng

Weekly Progress

Direction 1: Physical-Similarity-Based STS

- Redefine similarity based on shared physical effects of robot instructions.
- Construct a Physical Overlap Score using ALFRED metadata: actions, objects, and locations.
- Applications: Semantic Gap Analysis and SBERT Domain Adaptation.

Key Challenge

The metric is heuristic-based, raising concerns about its validity as an objective ground truth.

Additional Work This Week

- Exploration of an Alternative Research Direction: Explainability Analysis of Sentence-BERT in case direction 1 won't work due to it's metric problem

Made with GAMMA

Building and Analyzing the PSC Dataset

Creating a balanced dataset of instruction pairs annotated with PSC scores and examining its characteristics

Pair Generation: Generated 900 task pairs by sampling from all combinations, ensuring balanced representation across similarity levels

Balanced Sampling: Pairs are categorized into three buckets (high, mid, low similarity) based on their PSC score, with 300 pairs in each bucket

Dataset Content: Each pair includes: original instructions (sentence1, sentence2), psc_score, and detailed action_match, object_sim, location_sim values

Made with GAMMA

Understanding the ALFRED Dataset

Extracting and analyzing the core components of ALFRED tasks relevant to physical state changes and natural language instructions

Data Loading: Loaded 251 unique tasks each with a task_id

Key Information Extracted: Each traj_data.json provides:

- task_type: e.g., pick_heat_then_place_in_recep
- pddl_params: Physical parameters like object_target, parent_target
- turk_annotations.anns: Natural language instructions (e.g., "to cook a sliced tomato to throw it in the trash.")

Task Type Distribution: Demonstrated the variety and frequency of different task categories within the dataset

Made with GAMMA

Physical State Change STS (PSC-STS): Task-Oriented Semantic Textual Similarity for Robot Instruction

Made with GAMMA

Defining the Physical State Change (PSC) Score

Introducing a novel metric to quantify semantic similarity based on physical changes, crucial for robot understanding

Problem: Traditional linguistic similarity often fails to capture robot-relevant physical differences

Solution: PSC Score = w1·Action_Match + w2·Object_Sim + w3·Location_Sim

Action Match

Compares task_type (e.g., same pick operations = 0.5)

Object Similarity

Hierarchical similarity (same object = 1.0, same category = 0.7)

Location Similarity

Hierarchical similarity (same location = 1.0, same room = 0.5)

Made with GAMMA

Result 1 (Identical Tasks):

pick_heat_then_place_in_recep | Tomato → GarbageCan vs.

pick_heat_then_place_in_recep | Tomato → GarbageCan

identical task. PSC Score = 1.0 (Action=1.0, Object=1.0, Location=1.0)

Result 2 (Different Tasks):

pick_heat_then_place_in_recep | Tomato → GarbageCan vs.

pick_two_obj_and_place | SprayBottle → GarbageCan.

PSC Score = 0.5 (Action=0.5, Object=0.0, Location=1.0)

Made with GAMMA

📋 Samples (High Score):

	sentence1	sentence2	psc score
129	Put a frozen tomato in to garbage bin.	Slice a potato with a knife in order to throw it away.	0.731
194	Place a cooled mug on a coffee machine.	Put a clean mug under the coffee maker.	0.830
215	Put the green spray bottles on the toilet	Put a clean cloth on the back of the toilet.	0.731

📋 Samples (Low Score):

	sentence1	sentence2	psc score
0	Put the credit card in the drawer.	Grab the computer from the arm chair, turn off the floor lamp	0.00
1	Move a plunger to the cabinet.	Put two remotes on an ottoman.	0.17
2	Place a cooked slice of apple to the right of the yellow knife on the counter.	Put a bottle on the back of a newspaper.	0.17

Made with GAMMA

Baseline SBERT Score Calculation

- Loaded the pre-trained all-MiniLM-L6-v2 Sentence-BERT (SBERT) model
- Computed cosine similarity scores for all 900 instruction pairs in the dataset

Made with GAMMA

Semantic Gap Analysis

- Calculated Spearman's rank correlation coefficient (ρ) between SBERT baseline scores and PSC scores
- Identified 'false positives' (SBERT high, PSC low) and 'false negatives' (SBERT low, PSC high)
- Generated visualizations: scatter plots, SBERT distribution by PSC bucket, sensitivity to object changes

Made with GAMMA

Key Findings:

- Strong positive Spearman's ρ of approximately 0.73-0.75 observed
- Identified semantic gaps: 0 false positives and 2 false negatives

```
S1: place a microwaved tomato inside of the fridge
S2: Put cabbage in the refrigerator, use the knife to cut the cabbage
SBERT=0.282 | PSC=0.731 | Gap=-0.449
[Physical]: Tomato→Fridge vs Lettuce→Fridge

S1: to cook a sliced tomato to throw it in the trash.
S2: Put a warm potato in the sink.
SBERT=0.277 | PSC=0.736 | Gap=-0.459
[Physical]: Tomato→GarbageCan vs Potato→SinkBasin
```

Problem Statement: While SBERT tends to underestimate similarity by reacting to secondary verbs or objects in multi-step instructions, the current PSC metric focuses solely on primary physical parameters, resulting in inflated similarity scores that ignore the task's composite intent.

Proposed Solution: It is necessary to extend the framework into a "Multi-step Aware PSC Metric" that decomposes instructions into individual verb phrases, calculates PSC scores for each, and integrates the temporal sequence and logical dependencies of the entire task.

Made with GAMMA

Next steps:

Domain Adaptation (Fine-tuning)

- Split df_pairs dataset into training and evaluation sets
- Fine-tuned all-MiniLM-L6-v2 SBERT model using training data

Evaluation and Comparison of Spearman's ρ (between fine-tuned SBERT model and PSC)

Weight Sensitivity Analysis

- Perform grid search over different weight combinations for Action, Object, and Location components
- Test combinations of w1 (action), w2 (object), w3 (location) weights
- For each combination, calculated new PSC scores
- Assessed correlation with both baseline and fine-tuned SBERT models

Made with GAMMA

Interpreting Semantic Text Similarity: An Explainability Analysis of Sentence-BERT

From performance metrics to decision mechanisms — opening the "black box" of natural language processing models

EXPLAINABLE AI · SEMANTIC SIMILARITY · NLP RESEARCH

Core Research Philosophy

From Black Box to Transparency

Go beyond simple performance metrics to understand why the model predicts specific similarity scores.

Research Focus

Systematically analyze the decision-making mechanism of Sentence-BERT (SBERT) in semantic textual similarity tasks.

Made with GAMMA

Overall Research Workflow

Data Preparation

Load STS Benchmark and review gold scores

Baseline Construction

Use all-MiniLM-L6-v2 SBERT as benchmark

Performance Evaluation

Compute Pearson and Spearman correlations

Interpretability Analysis

Select high-error and high-similarity cases

Made with GAMMA

Core Analysis: Token-Level Alignment

Opening the “Black Box”

By computing the **pairwise cosine similarity** between word embedding vectors, we enable fine-grained token-level semantic alignment analysis. A **heatmap** is used to visually present the correspondence between the two sentences.

Case Example Sentence A: *"A man is playing a guitar."* Sentence B: *"A person is performing with a guitar."* **Finding:** man ↔ person (synonymous substitution) and playing ↔ performing (semantic equivalence) both show strong alignment signals.

Made with GAMMA

Error and Linguistic Phenomena Analysis

Systematically test the model's performance boundaries under four typical language challenges:

Synonym Generalization

Can the model recognize semantic equivalence under non-exact lexical matching? (e.g., man ↔ person)

Negation Logic

Can the model capture semantic contradictions? (e.g., *is playing* vs. *is not playing*)

Numerical Sensitivity

How capable is the model of perceiving fine-grained quantitative differences? (e.g., *two dogs* vs. *three dogs*)

Lexical Overlap Trap

When faced with sentence pairs that are lexically similar but semantically different, is the model's judgment influenced by surface-level lexical overlap?

Made with GAMMA

Core Research Questions

RQ1: Semantic Contribution

Which words or semantic relationships contribute most to the model's similarity judgments?

RQ2: Language Phenomena Capture

Can the model correctly handle negation, synonym replacement, and other complex language phenomena?

RQ3: Failure Scenario Analysis

In which specific scenarios does the model perform inaccurately? What are the underlying deeper causes?

Made with GAMMA

Q&A

Question 1

Objectivity of the Physical Overlap Score

Given that the "Physical Overlap Score" is derived from heuristic rules over task metadata rather than human annotations, can it truly be considered an objective form of ground truth?

Question 2

Research Direction Preference

Between fine-tuning SBERT and critically evaluating the validity of the Physical Overlap Score, which research direction would be more promising or impactful?

Made with GAMMA